

02-719 Week 1

Genomics and Epigenetics of the Brain

Aidan Jan

August 28, 2026

Logistics

- Office hours: Wed. 1-2 PM, GHC 7711
- Email: apfenning@cmu.edu

Motivation of Neurogenomics

Alzheimer's Disease

- More than 7 million Americans are currently living with Alzheimer's. (Projected to increase to 13 Million by 2050)
- 1 in 3 older adults dies with Alzheimer's or another dementia. It kills more than breast and prostate cancer combined.
- \$409 Billion dollars on Alzheimer's spendings in 2026, projected to be \$1 Trillion in 2050
- The lifetime risk for Alzheimer's at age 45 for women is 1 in 5, and for men it is 1 in 10. (Unknown reason)

Parkinson's Disease

- Lower prevalence than Alzheimer's (1 in 100 over age 60)
- 1M people in the United States, and 5M people worldwide affected

Addiction

- 46.3M people in the US have a substance use disorder in 2021.
- 105k drug overdose deaths per year

Major Depressive Disorder

- 13.8M - 15.4M adults affected
- One of the leading causes of suicide
- Costs employers \$78.7 Billion per year

Disease Categories

Most of these diseases fall into one of these following categories:

- No cure
- No effective long-term treatment (Alzheimer's, Parkinson's, other neurodegenerative diseases)
- Treatment comes with severe side effects (Bipolar, Major Depressive Disorder, ADHD, other psychiatric disorders)

Why are Brain Disorders So Hard to Treat?

The primary issue: What do you target when you treat a brain disorder?

- Target a specific brain region
- Specific cell types
- Specific hormones and neurotransmitters
- Specific proteins
- Specific section of DNA (gene therapy)
- Specific neural pathway

The problem is that modern medicines for brain diseases are like sledge hammers. One protein might influence the disease, but also regulate many other brain functions. Same goes with neurotransmitters, genes, etc. There isn't a single gene, protein, brain region, or biochemical pathway that equates to a single behavior.

Cellular Diversity of the Brain

The cellular diversity of the brain is overwhelming.

- Purkinje
- Microglia
- Astrocytes
- Bipolar
- Oligodendrocytes
- Schwann

Brief History of Neuroscience

- Santiago Ramon y Cajal (1852 - 1934): Traced neuron morphology and noticed there were different sub-types. Nobel prize winner.
- Cajal had a modern Zeiss microscope, which scientific discovery would not be possible without.
- Cajal's work was more "data-driven" than "hypothesis-driven". Rather than designing an experiment to test a specific theory of the brain, Cajal sought to characterize the diversity of neural cell types
- His work led to the neuron doctrine: "The brain is made of many different types of neurons."

Distinguishing Subtypes of Neurons

- Shape (Cytoarchitecture)
- Firing (Electrophysiology)
- Neurotransmitter (Glutamatergic vs. Gabaergic)
- Marker Genes (Somatostatin)
- Genomics

Single Neuron RNA-Seq measures genome-wide gene expression in single cells.

Epigenetics

There is an extraordinary diversity of neural cells within the brain that have different molecular and physiological properties. Differences in gene expression are caused by **epigenetic differences** in enhancers across neural cell types.

- There are differences in the genome sequence of enhancers that influence the evolution of complex traits (and human disease).
- Typical computational methods do not capture cell type-specific evolution at enhancer sequences.
- Machine learning has the potential to trace the evolution of neural cell type-specific enhancers across evolution.

Studying the Brain

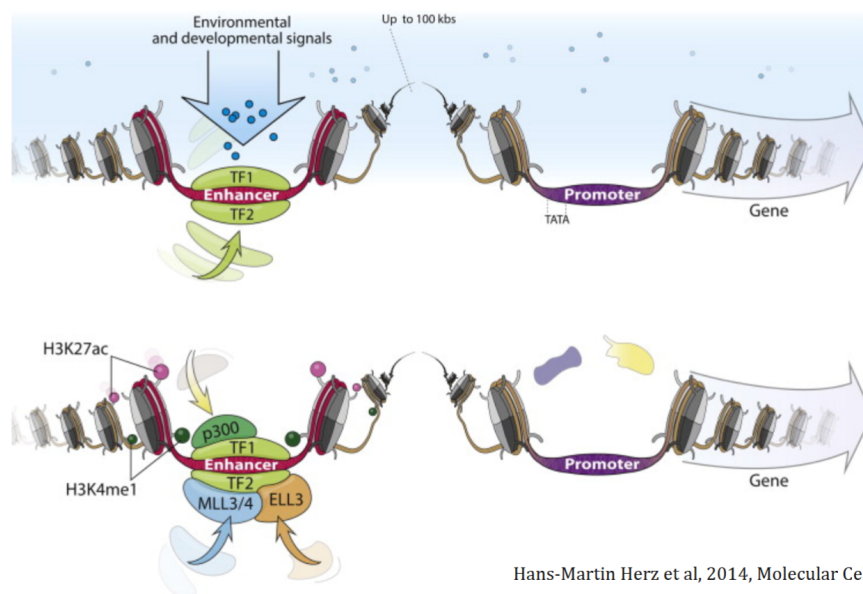
There are many challenges of studying the brain posed by cell types.

- Deep brain stimulation / optogenetics - How do you selectively stimulate the desired cell type in a given brain region
- Neurological / Psychiatric disorders - What cell type(s) are underlying the disorder? How do you target them?
- How do distinct subtypes of neurons form neural circuits for learning/memory?

Different Views of the Genome

Computer scientists see the genome as a sequence of A's, T's, C's, and G's, and focus on identifying specific DNA sequences that might correlate with diseases. However, this view often does not capture the true complexity of cell types since it dismisses epigenetics completely. Biologists not only see the genome, but also consider which parts might code for proteins, which might be transcriptionally active, which might be repressed. Epigenetics, enhancers, and other regulatory elements are considered.

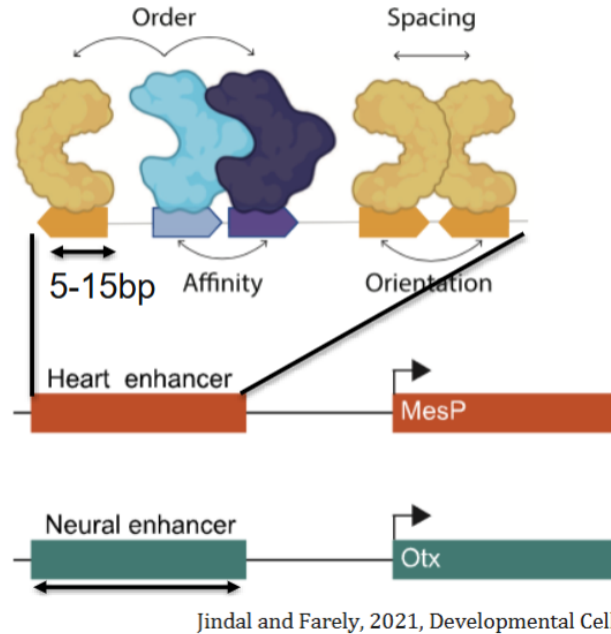
Epigenetic Mechanisms of Gene Regulation



Hans-Martin Herz et al, 2014, Molecular Cell

Different cell types might have the same gene, but regulated by different enhancers. A given gene might have anywhere between 2 to over 100 enhancers regulating it.

- The combination and spacing of **transcription factor binding sites** determines activation.
- Enhancers have a ‘grammar’ to be able to operate correctly.
 - Their spacing, orientation, and order all matter.
 - Proteins bind to enhancer sequences on the DNA. Two enhancers too close together or facing the wrong direction may cause the binding of one protein to inhibit the binding of the other protein.



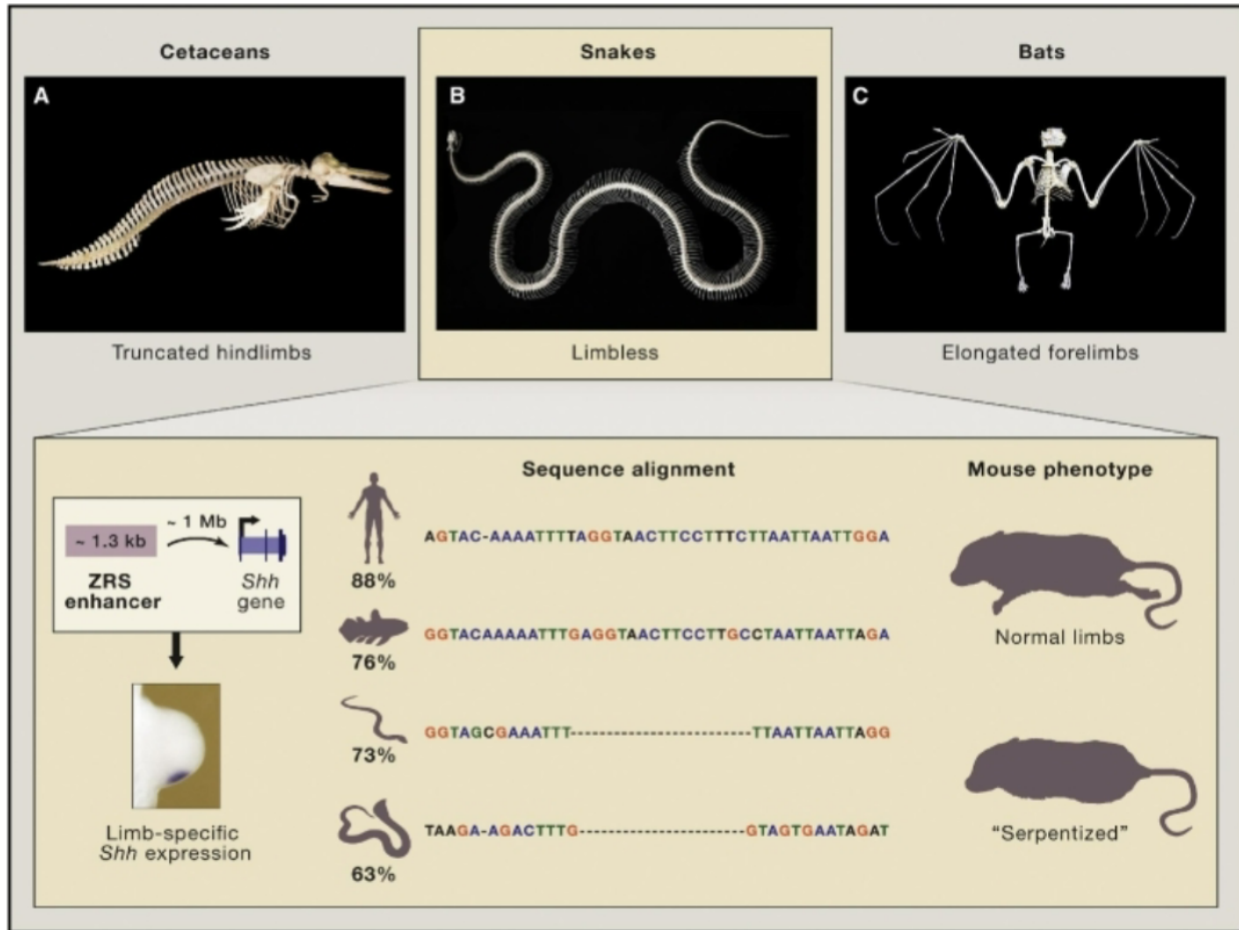
- There are hundreds of thousands of enhancers across the human brain (and millions across the body)
- Most of these enhancers are cell type-specific
- Promoter regions and the genes themselves are much less specific

Conservation through Evolution

Conservation methods are protein-centric.

- If something has been maintained over millions of years of evolution, then there must be evolutionary constraint selecting against mutations in these regions.
- In protein-coding regions, every third base pair has a higher rate of mutating than the first two due to the **third codon wobble**; when observing the codons for different amino acids, the identity of the amino acid depends less on the third base pair compared to the first two.
- Conservation can be measured by PhyloP / PhastCons. (Pollard et al., 2010)

Below is an example of how an enhancer might be linked to a physical trait.

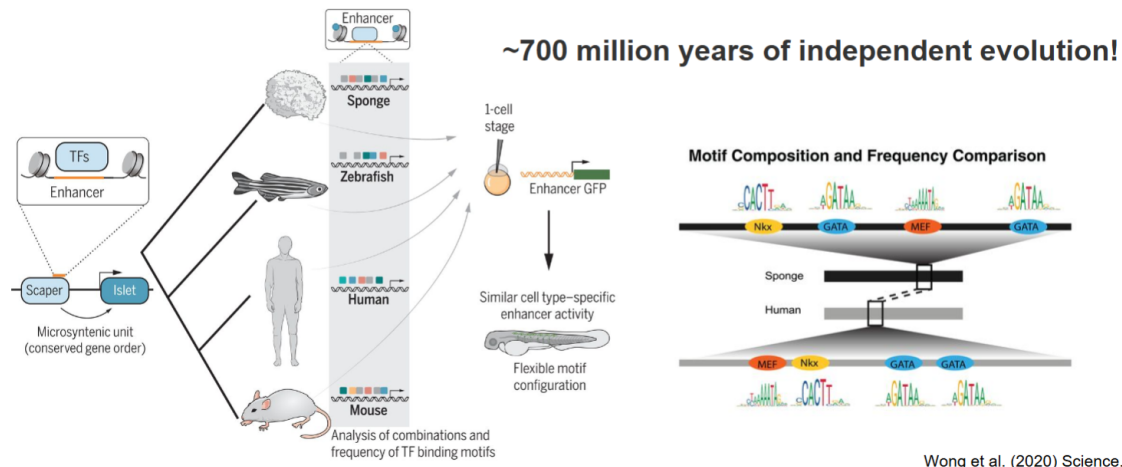


Cell 2016 167598-600DOI: (10.1016/j.cell.2016.10.007)

- The ZRS enhancer was discovered to be related to limb development when observing different organisms. Snakes have the enhancer sequence deleted.
- When scientists deleted the same sequence from mice DNA, they "Serpentized" the mice, causing their limbs to not develop at birth.

Alignment of Individual Nucleotides is not Conservation

Is Nucleotide-level conservation the exception or the rule?



- It turns out that nucleotide-level conservation is **not** the rule. Genes are selected for, not base pairs.
- In the different species on the diagram, they all share a Scaper enhancer which enhances a gene called Islet. It turns out that this combination is present in all the species listed, and their nucleotide motifs are conserved, albeit appearing in different orders.

Nucleotide alignment does not capture regulatory evolution.

- There is no strong difference in nucleotide-level evolution between enhancers that are conserved and divergent.

Machine Learning for Enhancer Prediction

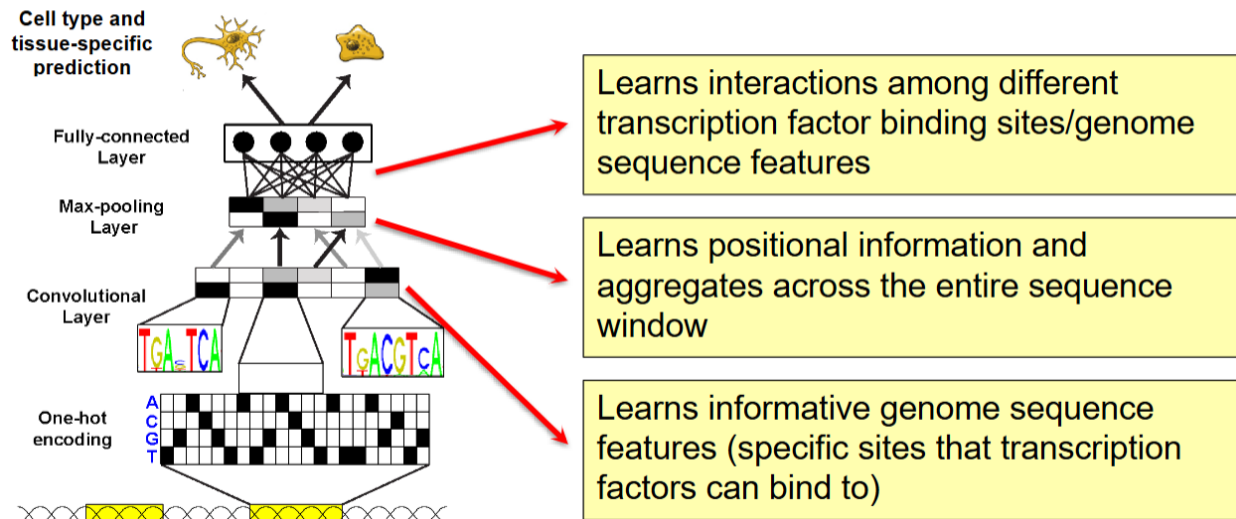
Studies have been done on training machine models on Motif scores, k-Mers, and raw sequences of nucleotides. It turns out that raw sequence has by far the best performance.

- Alternatively, use tissue-specific regulatory code.

How do we find Conserved Segments of the Genome?

- Convolutional neural network (deep learning) has revolutionized image analysis.
- We can adapt this method to make predictions of genome sequence function.

Convolutional neural network model linking genome sequence to enhancer activity

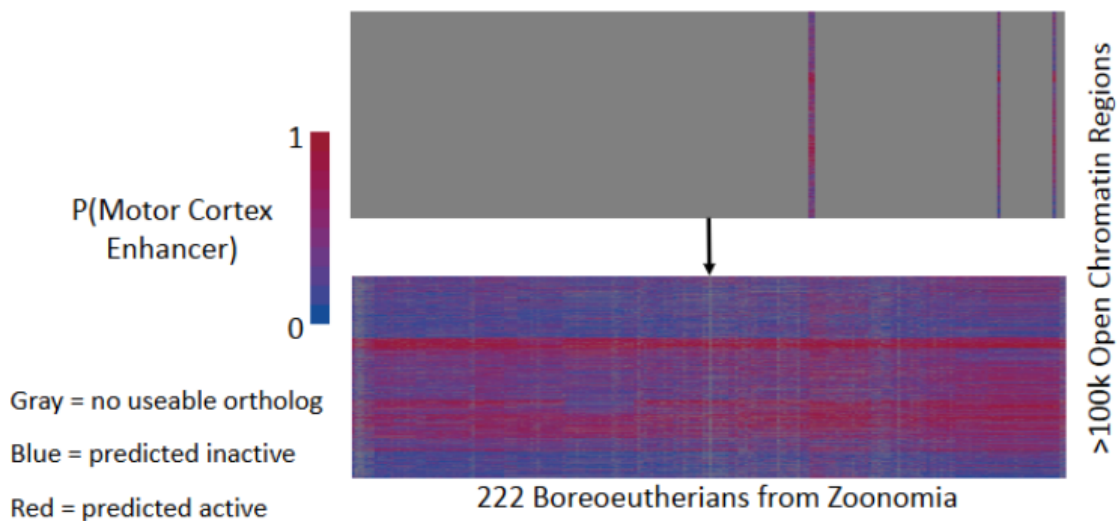


Alternatively, use tissue-specific regulatory code

- We can measure activity correlated with certain enhancer sequences in some species. . .
- and use it to predict activity in orthologous regions of other species.

The Zoonomia Consortium

- A series of papers sequencing DNA in brains (motor cortex) from multiple species, and predicting activity in motor cortex enhancers



- The Y-axis are the open chromatin regions.
- The X-axis are different species.
- The input is the top graph, where only a few species are sampled for enhancer activity.
- The output is the bottom graph, essentially extrapolating from the input species.
- The CNN models are accurate, with >80% validation accuracy.

The above project is called TACIT-ML (Tissue Aware Conservation Inference Through Machine Learning), and it outperforms previous models, such as PhyloP and PhastCons.

- TACIT (predicted open chromatin) does a better job at identifying which mutations are associated with human disease and what cells they function in.
- TACIT can be used to identify part of the genome associated with the evolution of complex traits

Brain Size has Convergently Evolved

There is an incredible diversity of mammalian brains.

- It turns out that when plotted on a logarithmic scale graph (both axes), body correlates linearly with brain weight